MYCAMPUSVIEW ACADEMY · ROBOTICS 101

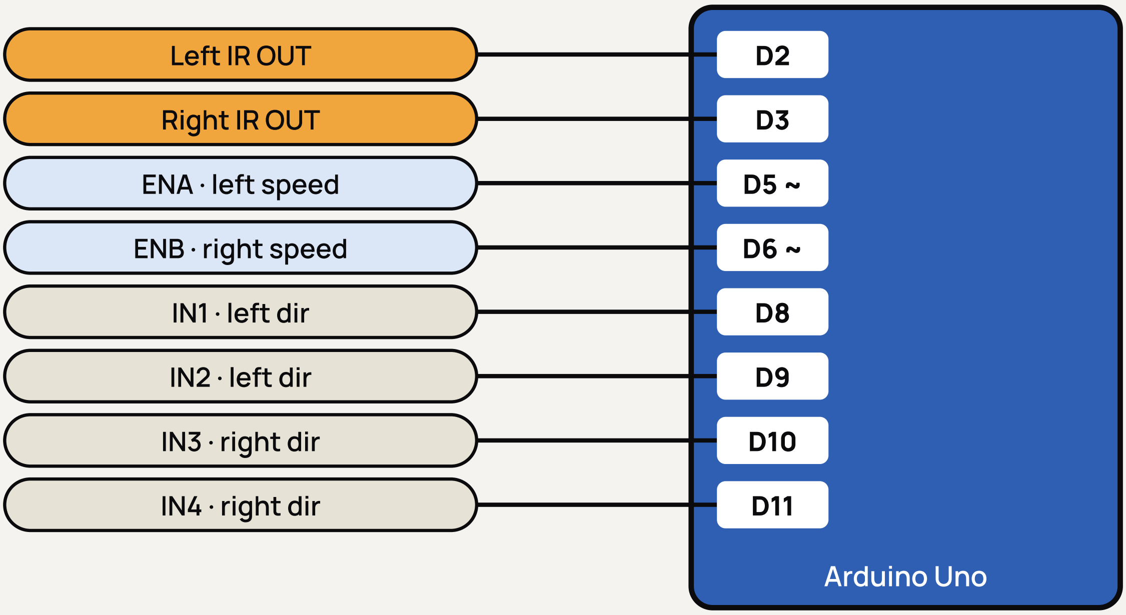
Pin connection reference

Print and keep it next to your robot

Signal	Arduino pin	Name in code	Type
Left IR sensor OUT	**D2**	`LEFT_IR`	Digital input
Right IR sensor OUT	**D3**	`RIGHT_IR`	Digital input
L298N ENA (left motor speed)	**D5**	`ENA`	PWM output (~)
L298N ENB (right motor speed)	**D6**	`ENB`	PWM output (~)
L298N IN1 (left motor direction)	**D8**	`IN1`	Digital output
L298N IN2 (left motor direction)	**D9**	`IN2`	Digital output
L298N IN3 (right motor direction)	**D10**	`IN3`	Digital output
L298N IN4 (right motor direction)	**D11**	`IN4`	Digital output

Power: each IR sensor VCC → 5V and GND → GND. Battery + → switch → L298N 12V; battery – → L298N GND; L298N GND → Arduino GND (common ground). Left motor → OUT1/OUT2 (motor A); right motor → OUT3/OUT4 (motor B).

Reference pin map


Left IR OUT
Right IR OUT
ENA · left speed
ENB · right speed
IN1 · left dir
IN2 · left dir
IN3 · right dir
IN4 · right dir
D2
D3
D5 ~
D6 ~
D8
D9
D10
D11
Arduino Uno

Motor direction

Robotics 101: Build a line-follower

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IN1 / IN3	IN2 / IN4	Motor
HIGH	LOW	Forward
LOW	HIGH	Backward
LOW	LOW	Stop

Speed: `analogWrite(ENA, 0–255)` for the left motor, `analogWrite(ENB, 0–255)` for the right. Motor A is the left motor; motor B is the right.